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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences  
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## Assessment Tool 1 — Short Answer Test

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Course: COMPUTER VISION

Course Code: ITA0518

### Q1. Camera image blurred due to improper sampling and effect on quality & correction

Ans:Sampling converts a continuous (analog) image into a discrete grid of pixels by measuring intensity at fixed spatial intervals. The sampling rate (spatial resolution) determines how much real-world detail is preserved.

- If the sampling rate is too low relative to the spatial-frequency content of the scene (violating the Nyquist criterion — sampling must occur at least twice the highest frequency present), fine detail and edges cannot be captured. Neighbouring samples then represent averaged intensities from a larger real-world area, which is perceived as blur.
- Insufficient sampling also causes aliasing — high-frequency patterns (fine textures, repetitive structures) are misrepresented as false, lower-frequency patterns.
- Net effect on quality: loss of fine detail, blocky/blurred appearance, jagged edges, moiré patterns.

#### Correction methods:

- Increase the sampling rate (use a higher-resolution sensor) so the Nyquist condition holds for the frequencies present.
- Apply an anti-aliasing (low-pass) filter before sampling, removing frequencies that cannot be captured at the given rate.
- Use proper reconstruction/interpolation (e.g. bicubic) when resampling, to smooth out blockiness from inadequate sampling.

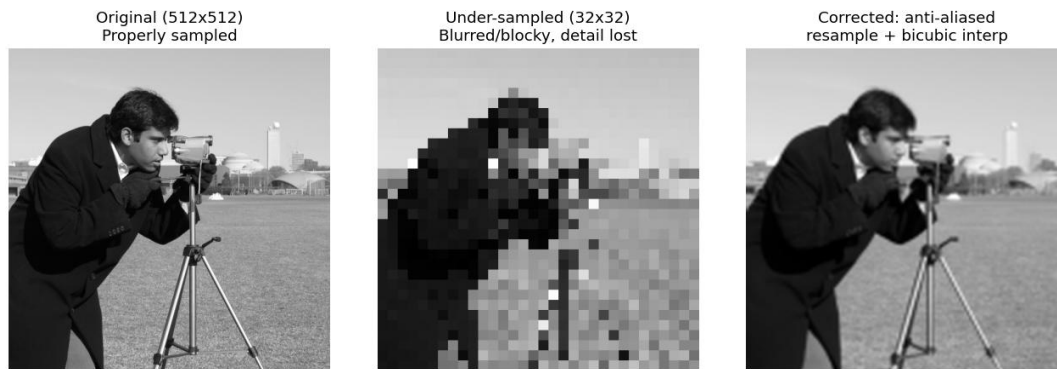
#### Python demonstration:

```
from skimage import data
from skimage.transform import resize

img = data.camera() / 255.0          # original, properly sampled 512x512 image

# Poor sampling: naive pixel skipping (no anti-alias filter), then upscale
low_naive = img[::16, ::16]           # 32x32 -- under-sampled
low_naive_up = resize(low_naive, img.shape, order=0, anti_aliasing=False)

# Correction: pre-filtered (anti-aliased) downsample + bicubic reconstruction
low_good = resize(img, (128, 128), anti_aliasing=True)
recon = resize(low_good, img.shape, order=3, anti_aliasing=True)
```



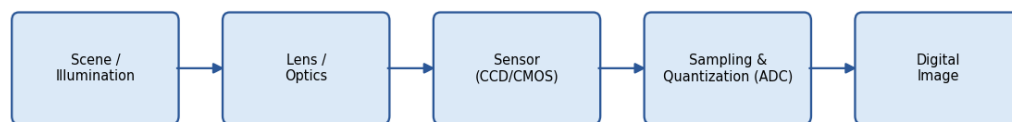
## Q2. Low-light scene — influence of image sensing & acquisition on the final digital image.

Ans:

### Effects of low light on this pipeline

- Reduced photon count: fewer photons strike each sensor element during the exposure interval, lowering the signal-to-noise ratio (SNR).
- Increased sensor gain/ISO: needed to boost the weak signal, but this also amplifies thermal and shot noise, producing a grainy image.
- Longer exposure time: gathers more photons but risks motion blur if the scene or camera moves.
- Reduced dynamic range: dark regions may fall below the sensor's detectable threshold (crushed shadow detail); quantization banding becomes more visible in smooth gradients since fewer effective levels are usable.
- Colour/tone shifts: white balance and colour accuracy degrade when insufficient light reaches all colour channels.

In short, acquisition conditions directly determine the noise level, contrast, dynamic range and sharpness of the resulting digital image. Mitigations include larger photosites/cooled sensors, multi-frame averaging, and adaptive exposure/gain control.



## Q3. Aliasing artifacts cause via sampling/quantization & corrective approach

Ans:

Cause: Aliasing is primarily a sampling problem, not a quantization one. It occurs when the scene contains spatial-frequency content higher than half the sampling rate (violating the Nyquist–Shannon theorem). These frequencies fold back and appear as false, lower-frequency patterns — moiré fringes, jagged edges, or repeated patterns — especially on fine repetitive textures (fabric weaves, blinds, etc.). Quantization contributes a related but distinct artifact — false contouring — when too few intensity levels represent a smooth gradient. This is caused by inadequate intensity (gray-level) resolution, not by the spatial sampling rate, and should not be confused with aliasing.

### Corrective approach

- Apply a low-pass (anti-aliasing) filter to the signal before sampling, to suppress frequencies above the Nyquist limit.
- Increase the sampling rate (higher sensor resolution) so the Nyquist frequency exceeds the scene's highest frequency.
- In software, blur (e.g. Gaussian smoothing) the image before downsampling to a lower resolution.

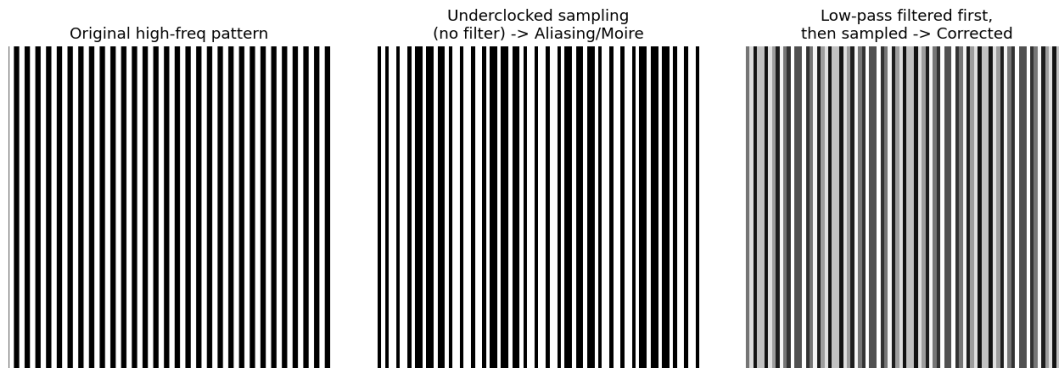
### Python demonstration

```
import numpy as np
from skimage.filters import gaussian

# High-frequency stripe pattern
pattern = (np.sin(X) > 0).astype(float)      # X from a fine meshgrid

aliased = pattern[::7, ::7]                  # naive sampling -> moire/aliasing

blurred = gaussian(pattern, sigma=3)          # anti-aliasing (low-pass) filter first
corrected = blurred[::7, ::7]                # then sample -> aliasing removed
```



## Q4. Pixel resolution vs. intensity resolution and impact in medical imaging

Ans:

### Pixel (spatial) resolution

The number of pixels used to represent the image (e.g.,  $512 \times 512$ ), determining the smallest resolvable spatial detail. Higher pixel resolution lets fine anatomical structures — small lesions, vessel boundaries, fracture lines — be distinguished clearly.

### Intensity (gray-level / radiometric) resolution

The number of distinct intensity levels used to represent brightness, tied to bit-depth (8-bit = 256 levels, 12-bit = 4096 levels). Higher intensity resolution captures subtle contrast differences — e.g., differentiating grey matter from white matter in an MRI — without visible banding.

### Impact in medical imaging

- Low pixel resolution → structures blur together or small abnormalities go undetected, even with fine contrast.
- Low intensity resolution → contrast steps appear as artificial bands (false contouring), hiding subtle pathology even with fine spatial detail.
- Both must be matched to the clinical task: mammography needs very high spatial resolution for micro-calcifications; CT/MRI often need high intensity resolution (12–16 bit) to differentiate soft-tissue types.

### Python demonstration

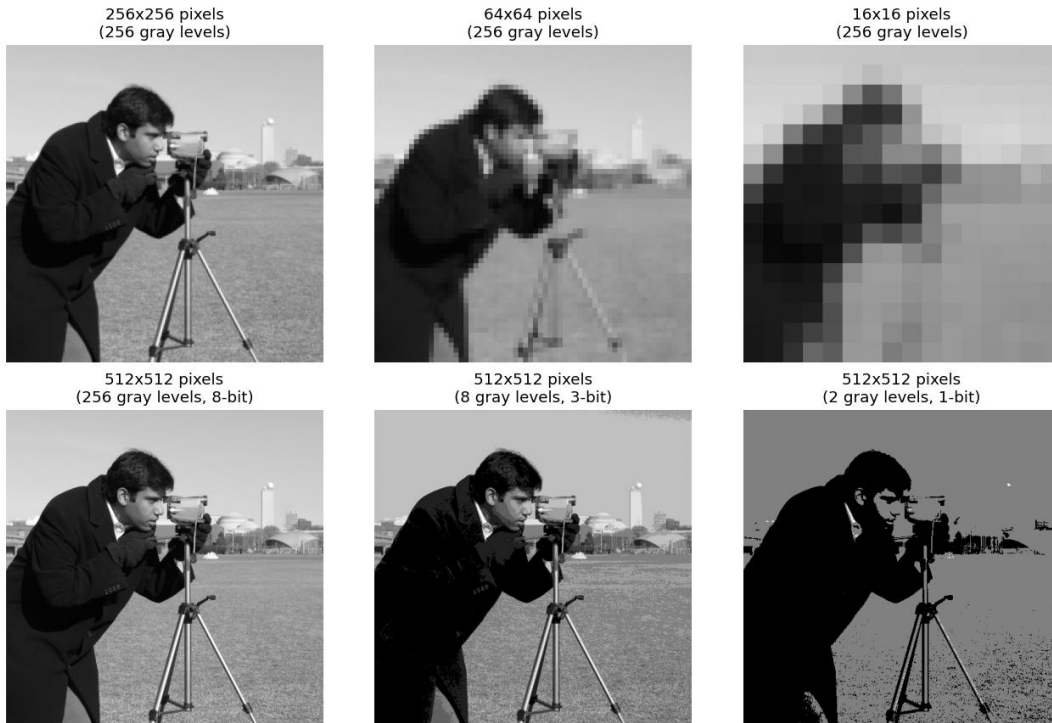
```

from skimage.transform import resize
import numpy as np

# Row 1: vary PIXEL resolution, fixed 256 gray levels
for s in [256, 64, 16]:
    small = resize(img, (s, s), anti_aliasing=True)

# Row 2: fixed 512x512 pixels, vary INTENSITY resolution (bit depth)
for levels in [256, 8, 2]:
    quantized = np.floor(img * levels) / levels

```



## Q5. Real-time surveillance system and classifying the level of Computer Vision

Ans:

Computer Vision tasks are commonly classified into three levels:

- Low-level: pixel-level operations — noise removal, edge detection, contrast enhancement, filtering. Fast; no semantic understanding.
- Mid-level: grouping/structure — segmentation, feature extraction (SIFT/HOG), motion/optical flow, boundary/contour detection.
- High-level: semantic interpretation — object recognition/classification, identity tracking, scene understanding, decision-making.

A real-time surveillance system spans mostly low-level (frame differencing/noise filtering for speed) and mid-level (foreground segmentation, motion detection, blob/feature extraction) processing, with a high-level component for recognition/classification and alert decisions (e.g., person vs. vehicle, unauthorized access).

Justification: the real-time constraint pushes emphasis toward low/mid-level processing, since these must run at video frame rate; high-level recognition may run at a reduced rate or asynchronously (e.g., every Nth frame) due to its higher computational cost.



## Q6. Image formation models improving image quality in autonomous driving

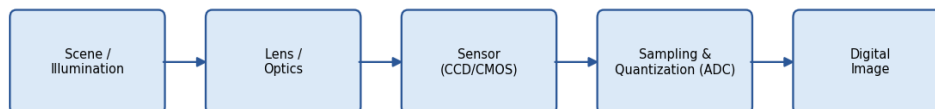
Ans:

An image formation model mathematically describes how a 3D scene (illumination, geometry, reflectance) is projected into a 2D digital image, e.g.  $I(x,y) = i(x,y) \cdot r(x,y)$  (illumination  $\times$  reflectance), combined with a geometric camera model (pinhole projection with intrinsic/extrinsic parameters) and a radiometric/sensor response model.

### Applications in autonomous driving

- Camera calibration (from the geometric formation model) maps image pixels to real-world coordinates — essential for distance/depth estimation and object localization.
- Illumination models help correct varying lighting (shadows, glare, night driving), improving robustness of downstream detection.
- Lens distortion models (radial/tangential) allow the system to undistort images for accurate lane detection and boundary estimation.
- Stereo/geometric formation models enable depth perception (stereo vision, structure-from-motion) needed for obstacle avoidance.
- Noise/sensor models guide appropriate denoising so detection/segmentation networks receive cleaner input.

Overall, image formation models provide the mathematical foundation to correct, calibrate, and interpret raw sensor data — directly improving detection accuracy, depth estimation, and safety.



## Q7. Role of quantization when converting a grayscale image into a digital image

Ans:

Quantization maps a continuous range of intensity values to a finite set of discrete levels (e.g., 256 levels for 8-bit). It is the second key step (after sampling) in digitizing an image, determining how finely brightness variations are represented.

### Impact of insufficient quantization levels

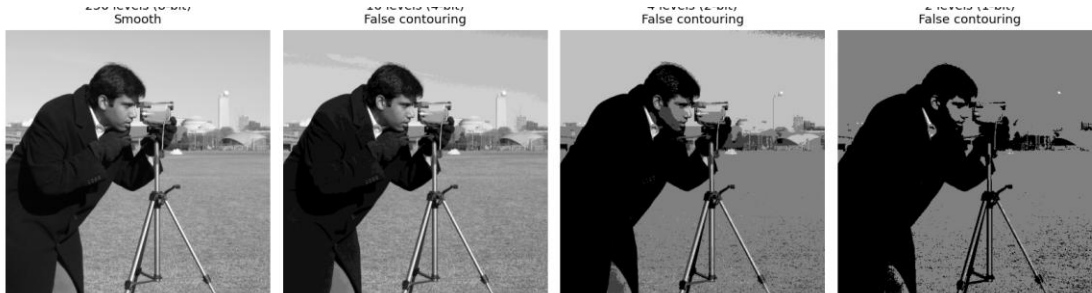
- False contouring: smooth gradients (sky, skin tones) appear as visible, artificial bands/steps instead of smooth transitions.

- Loss of subtle detail: fine contrast differences used to convey shape become indistinguishable.
- More levels (higher bit depth) → smoother tonal transitions and more accurate photometric detail, at the cost of larger storage size.

### Python demonstration

```
import numpy as np

for levels in [256, 16, 4, 2]:
    quantized = np.floor(img * levels) / levels    # reduce gray levels
```



## Q8. Noisy image from a sensor and how acquisition contributes noise & how to minimize it

Ans:

### Noise sources introduced during acquisition

- Photon (shot) noise: statistical fluctuation in photon count reaching a sensor element — pronounced in low light.
- Thermal (dark current) noise: random electron generation due to sensor heat, significant in long exposures.
- Read/amplifier noise: electronic noise from converting/amplifying the analog signal, especially at high ISO/gain.
- Quantization noise: rounding error when the ADC maps continuous voltage to discrete digital levels.
- Transmission/impulse noise: bit errors during transfer, appearing as salt-and-pepper noise.

### Minimization strategies

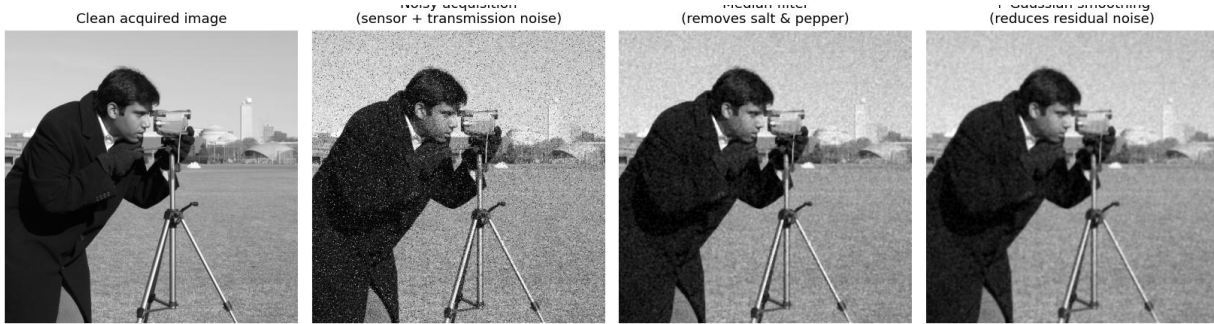
- Hardware: cooled sensors, larger photosites/better optics, lower ISO/gain, multi-frame averaging (frame stacking).
- Software: Gaussian smoothing (reduces random Gaussian noise, can blur edges); median filtering (removes salt-and-pepper noise while preserving edges); advanced methods (bilateral filter, non-local means, wavelet denoising) preserve edges while removing noise.

### Python demonstration

```
from skimage.util import random_noise
from skimage.filters import median, gaussian
from skimage.morphology import disk

noisy = random_noise(img, mode='gaussian', var=0.01)
noisy = random_noise(noisy, mode='s&p', amount=0.03)    # simulate sensor + transmission noise

median_denoised = median(noisy, disk(2))                # removes salt & pepper
gauss_denoised = gaussian(median_denoised, sigma=1)    # smooths residual noise
```

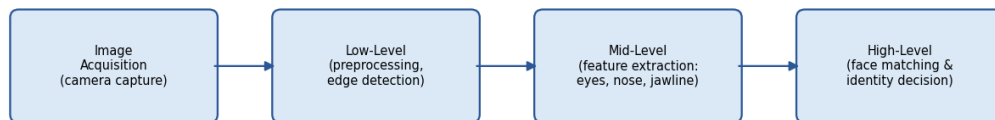


## Q9. Facial recognition and levels of Computer Vision from acquisition to decision

Ans:

- Image acquisition: the camera captures the face under varying pose/lighting; sensing and initial digitization occur here.
- Low-level processing: preprocessing — noise removal, illumination/contrast normalization, edge detection — to prepare the raw image.
- Mid-level processing: face detection (locating the face region) and feature extraction — key landmarks (eyes, nose, mouth, jawline) forming a feature descriptor/embedding of the face's geometry and texture.
- High-level processing: matching the extracted representation against a database (recognition/verification) and making a decision — identity match/no-match, access granted/denied.

This layered flow shows how raw pixel data is progressively transformed into meaningful, decision-ready information.



## Q10. Low-resolution image fails to detect objects and effect of sampling & image formation on object visibility

Ans:

Object visibility/detectability depends on whether the imaging pipeline (formation + sampling) preserves enough spatial detail and contrast for that object.

### Sampling effect

If the sensor's sampling rate (pixel density) is too low relative to the object's size/spatial frequency, an object spanning only a few pixels cannot be resolved — it may vanish or merge with the background, directly violating the Nyquist requirement for that object's spatial-frequency content.

### Image formation effect

The formation model — lens blur (point spread function), depth of field, and the illumination–reflectance interaction — determines how faithfully the projected 2D image preserves the object's edges/contrast before sampling even occurs. Poor focus, motion blur, or low contrast at the formation stage further reduces the information available, compounding the sampling limitation.

### Combined impact & correction

Even a well-designed detection algorithm cannot recover information that was never captured. Correction: increase sensor resolution/sampling rate relative to the smallest object of interest, use anti-aliasing filters, ensure correct focus/exposure during formation, and where hardware limits exist, apply super-resolution/careful interpolation (which enhances apparent detail but cannot invent lost real information).

Original (512x512)  
Properly sampled



Under-sampled (32x32)  
Blurred/blocky, detail lost



Corrected: anti-aliased  
resample + bicubic interp

